Case Study / Case Report

Emerging Clinical Significance of *Ochrobactrum anthropi* Septicemia in a Multimorbid Intensive Care Unit Patient: A Clinical Case Report

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Abstract
Ochrobactrum anthropi is a rare opportunistic Gram-negative, non-fermenting bacillus increasingly recognized as a cause of nosocomial infections, particularly in critically ill patients with indwelling catheters. Its ability to form biofilms and adhere to synthetic materials makes catheter-related bloodstream infections difficult to treat. We report a case of persistent *O. anthropi* bacteraemia in a hospitalized patient with a central venous catheter who presented with recurrent fever. Blood culture identified *O. anthropi* with susceptibility to carbapenems and resistance to several commonly used antibiotics. The patient was initially treated with piperacillin-tazobactam and later escalated to meropenem based on culture sensitivity. Despite partial clinical improvement, repeat blood cultures remained positive. Removal of the central venous catheter resulted in sterile repeat cultures, complete resolution of fever, and significant clinical recovery, highlighting the importance of prompt source control.

Key Words: Antimicrobial resistance, Biofilm formation, Central venous catheter, *Ochrobactrum anthropi* bacteraemia, Opportunistic nosocomial infection, Source control

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Editorial note: The corresponding author's personal home address and personal e-mail given in the submitted manuscript have been replaced with the institutional affiliation, per house policy of not publishing personal contact details. Two references appear non-standard and are flagged in place below: Ref. 1's title ("Infection: the first case report in Malaysia") does not match its cited source (a microbiology textbook chapter), and Ref. 3 is a bare website URL without author/title/access-format structure — both need author verification before this is publication-ready.

I. Introduction

Ochrobactrum anthropi is a Gram-negative, non-fermenting bacillus increasingly recognized as an opportunistic cause of healthcare-associated bloodstream infections. Although generally considered a low-virulence organism, it can cause clinically significant infection in critically ill patients and those with indwelling intravascular devices. Its ability to persist on foreign surfaces and its variable antimicrobial susceptibility may contribute to persistent bacteraemia and therapeutic challenges. We report a case of persistent *O. anthropi* bacteraemia in a multimorbid critically ill patient with a central venous catheter, in whom bacteraemia persisted despite antimicrobial therapy and resolved following catheter removal. The case highlights the importance of recognizing an indwelling catheter as a potential source and the role of prompt source control in management.

II. Case Report

A sixty-three-year-old male with a central venous catheter in situ along the right internal jugular vein presented to the emergency department with shortness of breath (mMRC grade 4) and confusion for 2 days. He had a significant medical history of COPD, hypertension, type 2 diabetes mellitus, and chronic kidney disease, and a surgical history of permanent pacemaker implantation in 2017. He had previously been admitted for dyselectrolytaemia 15 days earlier. On general physical examination, the patient was disoriented and febrile (temperature 101°F). Vitals were BP 156/90 mmHg, PR 90 bpm, SpO_2 86%. He had bilateral pedal oedema, warm and moist extremities, and bilateral crepitations on auscultation. The patient was managed in the intensive care unit (ICU). Blood samples for culture were collected before antibiotics were administered, and intravenous piperacillin-tazobactam was started.

Haematological investigation revealed haemoglobin 8.9 g/dL, leukocytosis (total leukocyte count 19,900/mm^3, 93% neutrophils, with toxic granules), and platelet count 95,000/mm^3. ESR was 36 mm/hour and CRP was positive. Renal function tests showed sodium 134 mmol/L, potassium 4.7 mmol/L, creatinine 3.3 mg/dL, urea 89 mg/dL, and uric acid 10.8 mg/dL. Urinalysis showed albuminuria with 20–21 pus cells/hpf. Chest X-ray (PA view) showed bilateral patchy opacities, and echocardiography showed severe pulmonary hypertension.

Blood cultures, processed on the automated BacT/ALERT system, identified *Ochrobactrum anthropi*. Empirical piperacillin-tazobactam (3.375 g IV every 6 h, extended infusion) was escalated to meropenem (1 g IV every 12 h) for seven days once sensitivity confirmed susceptibility to carbapenems. Although the patient showed clinical improvement with reduced symptoms and fewer febrile episodes, intermittent fever persisted. A repeat blood culture again showed persistent growth of *O. anthropi*, indicating ongoing catheter-related bacteraemia despite appropriate antimicrobial therapy. Given the organism's known biofilm-forming ability and persistent device colonization, the central venous catheter was removed. Following removal, repeat blood cultures became sterile, the bacteraemia resolved completely, and the patient became afebrile with marked clinical improvement.

III. Discussion

Ochrobactrum anthropi infection (reclassified in some literature as *Brucella anthropi*) is a Gram-negative, motile, non-fastidious, non-fermenting bacillus with strict oxidative metabolism belonging to the family Brucellaceae[1,2]. Although considered a low-virulence organism, its clinical significance lies in its ability to form biofilms and survive in antiseptic solutions, making it an important opportunistic nosocomial pathogen.

The organism is commonly isolated from hospital water sources and contaminated medical environments, including normal saline, antiseptic solutions, dialysis fluids, and indwelling medical devices such as central venous catheters and drainage tubes. Its strong adherence to foreign bodies and synthetic surfaces contributes significantly to persistent hospital-acquired infections. Beyond bacteraemia, other reported manifestations include infective endocarditis, meningitis, peritonitis, septic arthritis, and soft tissue infections[3].

Several studies have reported *O. anthropi* bacteraemia as an important cause of catheter-related bloodstream infection, particularly in immunocompromised individuals and patients with prolonged indwelling catheter use. Yu et al. analysed 15 cases of *O. anthropi* bacteraemia, portraying the organism's opportunistic nature[4]. Cieslak et al. identified *O. anthropi* as the cause of catheter-associated bacteraemia in a three-year-old child with retinoblastoma[5]. Arora et al. reported a case of *O. anthropi* septicaemia in an elderly cardiac-care-unit patient, with contaminated hospital water implicated as the likely source[6]. Braun et al. described *O. anthropi*-induced endophthalmitis after cataract surgery, necessitating intraocular lens removal[7].

These studies emphasize *O. anthropi*'s ability to cause serious infections in vulnerable patients and highlight the need for strict aseptic practices. Our isolate demonstrated resistance to multiple antibiotics — including colistin, amoxicillin-clavulanate, piperacillin-tazobactam, and cephalosporins — while remaining susceptible to imipenem, meropenem, and trimethoprim-sulfamethoxazole.

In our case, the central venous catheter likely served as the portal of entry for this opportunistic pathogen, consistent with previously reported cases where catheter colonization played a central role in infection development. Despite its relatively low virulence, *O. anthropi* demonstrates significant antimicrobial resistance; it is frequently resistant to penicillin, ampicillin, beta-lactam/beta-lactamase inhibitor combinations, chloramphenicol, tetracycline, and colistin, but often remains susceptible to fluoroquinolones, aminoglycosides, trimethoprim-sulfamethoxazole, and carbapenems[8].

This case reinforces the importance of source control: catheter removal was ultimately essential for definitive resolution, in addition to appropriate antibiotic therapy. Effective management of *O. anthropi* infections requires prompt identification, often aided by automated culture systems, given the organism's frequent misidentification as *Pseudomonas* or *Brucella* species[2].

This case emphasizes that even low-virulence organisms such as *O. anthropi* can cause persistent, clinically significant bloodstream infections, particularly in hospitalized and immunocompromised patients with indwelling devices. Early identification, appropriate antimicrobial selection, and timely removal of infected catheters remain the cornerstone of successful management.

IV. Conclusion

This case highlights that *Ochrobactrum anthropi* is an emerging nosocomial pathogen that deserves significant medical attention because of its wide environmental distribution and clinical relevance, particularly in immunocompromised patients and those with indwelling catheters or prolonged ICU stay. Its ability to survive in hospital environments and contaminate medical devices increases the risk of healthcare-associated infections. This case also emphasizes the importance of strict infection control practices, prompt culture collection before initiating antibiotics, antimicrobial stewardship, and susceptibility-guided targeted therapy in achieving successful outcomes with unusual, multidrug-resistant nosocomial pathogens.

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